

# Interpret Division of Fractions

Flagship

Lesson 2-1-flagship

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## EVIDENCE LAB MISSION

### **| The Fraction Vault**

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact  $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$12 \div 3 = 4$$

*In  $3 \div 1/4 = 12$ , the dividend is 3 — it is the total being split*

### Dividend

Write the definition:

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$$12 \div 3 = 4$$

*In  $3 \div 1/4 = 12$ , the divisor is  $1/4$  — it is the size of each piece*

### Divisor

Write the definition:

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$$12 \div 3 = 4$$

*In  $3 \div 1/4 = 12$ , the quotient is 12 — there are 12 quarter-size pieces in 3*

### Quotient

Write the definition:

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$$3/4 \rightarrow 4/3$$

flip top and bottom

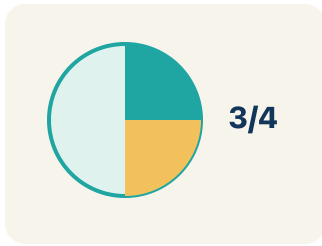
*The reciprocal of  $1/4$  is  $4/1 = 4$ . Multiplying by the reciprocal gives the same result as dividing.*

### Reciprocal

Write the definition:

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*$\frac{1}{2}$ ,  $\frac{1}{3}$ ,  $\frac{1}{4}$ ,  $\frac{1}{5}$  — each represents one equal part of a whole*

### **Unit fraction**

**Write the definition:**

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can use models to interpret what it means to divide by a fraction.**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Quotient

Reciprocal

Unit fraction

**1** In  $6 \div 1/2$ , the number 6 being divided is the \_\_\_\_\_.

**2** In  $6 \div 1/2$ , the number  $1/2$  we divide by is the \_\_\_\_\_.

**3** The answer to a division problem is the \_\_\_\_\_.

**4** The fraction you get by flipping the numerator and denominator is the \_\_\_\_\_.

**5** A fraction with a numerator of 1, like  $1/5$ , is a \_\_\_\_\_.



### Watch & Try — Worked Examples

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** What does  $3 \div \frac{1}{4}$  mean?

- A. How many  $\frac{1}{4}$ -size pieces fit into 3
- B. 3 groups of  $\frac{1}{4}$
- C. 3 minus  $\frac{1}{4}$
- D.  $\frac{1}{4}$  of 3

**Step 1**  $3 \div \frac{1}{4}$  asks: how many  $\frac{1}{4}$ -size pieces fit into 3?

**Step 2** The answer is 12, because each whole has 4 fourths, and  $3 \times 4 = 12$ .

 **Answer:** A. How many  $\frac{1}{4}$ -size pieces fit into 3

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** Which expression means 'how many  $\frac{1}{3}$ -size pieces are in 2'?

- A.  $2 \div \frac{1}{3}$
- B.  $2 \times \frac{1}{3}$
- C.  $\frac{1}{3} \div 2$
- D.  $2 + \frac{1}{3}$

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** A rope is 4 feet long. How many  $\frac{1}{2}$ -foot pieces can be cut from it?

- A. 8 pieces
- B. 2 pieces
- C. 4 pieces
- D.  $\frac{1}{2}$  piece

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Clue 2 — A roll of evidence tape is 8 feet long. The lab marks it into  $\frac{1}{4}$ -foot sections. How many sections are made, and which expression proves it?**

- A.  $8 \div \frac{1}{4} = 32$  sections
- B.  $8 \times \frac{1}{4} = 2$  sections
- C.  $\frac{1}{4} \div 8 = \frac{1}{32}$  section
- D.  $8 + \frac{1}{4} = 8 \frac{1}{4}$  sections

Show your work:

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**2. Clue 5 — Detective Reyes pours 5 quarts of evidence solution into beakers that each hold  $\frac{1}{3}$  quart. She wants to know how many beakers she fills. What does the quotient of  $5 \div \frac{1}{3}$  tell her in this story?**

- A. The number of  $\frac{1}{3}$ -quart beakers she can fill, which is 15
- B. The number of quarts in one beaker, which is  $\frac{1}{3}$
- C. How much solution is left over, which is 5
- D. How many quarts each beaker loses, which is 3

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Dividing 6 by  $\frac{1}{2}$  gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ( $\frac{1}{2}$ ) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.**

*Sentence starter: When I divide 6 by 2, I am \_\_\_\_\_. When I divide 6 by  $\frac{1}{2}$ , I am \_\_\_\_\_. The answer is bigger because \_\_\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**What does  $5 \div \frac{1}{4}$  mean, and what is the quotient?**

- A. How many  $\frac{1}{4}$ -size pieces fit into 5; quotient is 20
- B. 5 groups of  $\frac{1}{4}$ ; quotient is  $\frac{5}{4}$
- C.  $\frac{1}{4}$  of 5; quotient is  $\frac{5}{4}$
- D. 5 minus  $\frac{1}{4}$ ; quotient is  $4\frac{3}{4}$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Quotient
4. **Notes 4:** Reciprocal
5. **Notes 5:** Unit fraction
6. **We Try (notes):** A.  $2 \div 1/3$  — *Finding how many  $1/3$ -size pieces fit into 2 is represented by  $2 \div 1/3$ . The answer is 6.*
7. **You Try (notes):** A. 8 pieces —  $4 \div 1/2 = 4 \times 2 = 8$  pieces. *Each whole foot contains 2 halves, and  $4 \times 2 = 8$ .*
8. **Try It 1:** A.  $8 \div 1/4 = 32$  sections — *Each foot holds 4 fourths, so 8 feet hold  $8 \times 4 = 32$  sections. The matching expression is  $8 \div 1/4 = 32$ .*
9. **Try It 2:** A. The number of  $1/3$ -quart beakers she can fill, which is 15 —  $5 \div 1/3$  asks how many  $1/3$ -quart beakers fit into 5 quarts. *Each quart holds 3 thirds, so  $5 \times 3 = 15$  beakers.*
10. **Exit Ticket:** A. How many  $1/4$ -size pieces fit into 5; quotient is 20 —  $5 \div 1/4$  means 'how many  $1/4$ -size pieces fit into 5.' *Since each whole has 4 fourths,  $5 \times 4 = 20$ .*